



BIO-INSPIRED AI & WEARABLE TECHNOLOGY Enhancing Human Health & Performance

PRESENTED BY: MOHAMED ABDULLAH | MARWAN ZAKARIA | MAHMOUD MOHAMED
FACULTY OF SCIENCE, DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
DATE: APRIL 29, 2026

WHAT THIS PRESENTATION COVERS

**WHAT IS
BIO-INSPIRED
AI?**



**WEARABLE
TECHNOLOGY**



**HEALTH &
SPORTS
APPLICATIONS**



**FUTURE
IMPLICATIONS**



What if your **body** could
warn you... before it's too
late ?





WHAT IS BIO- INSPIRED AI ?

Content :

A type of artificial intelligence inspired by **natural systems**

Mimics how the human brain thinks and learns

Uses principles **of evolution** and natural behavior

Helps **solve** complex real-world problems **efficiently**



WHAT IS WEARABLES TECHNOLOGY ?

- **Content :**
 - Devices **worn** on the body
 - **Collect** real-time health data
 - **Equipped** with sensors (heart rate, BP, BS , Sleep)
 - Connected with AI for **smart analysis**



AI & *SLEEP* *Monitoring* *g*

Content:

Tracks sleep stages
(light, deep, REM)

Monitors heart rate
during sleep

Detects sleep disorders

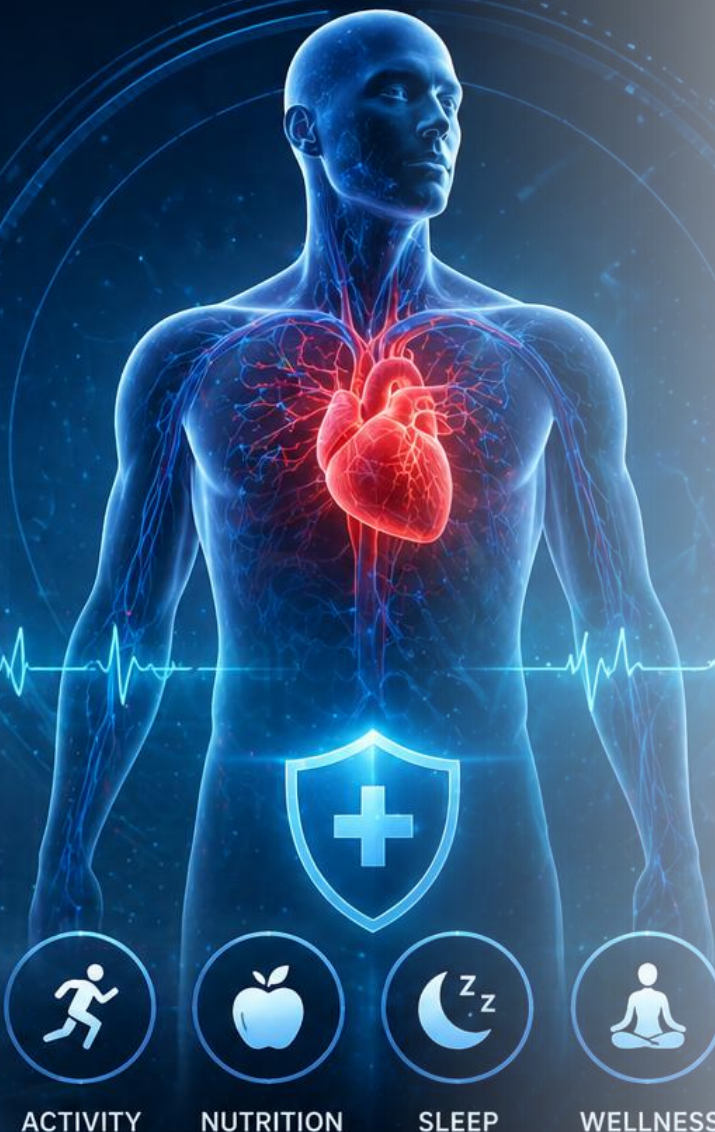
Improves recovery and
performance



BLOOD PRESSURE

Monitoring your pressure
for a healthy heart

120/80
mmHg



ACTIVITY

NUTRITION

SLEEP

WELLNESS

AI-POWERED INSIGHTS FOR A
HEALTHIER YOU

AI & Blood Pressure Monitoring

• Content :

- Continuous monitoring using wearables
- Detects abnormal patterns
- Sends real-time alerts
- Helps prevent heart diseases



BLOOD SUGAR

AI-POWERED MONITORING
Smart insights for better health



MONITOR

Tracks blood glucose levels in real-time



PREDICT

AI predicts highs and lows before they happen



ALERT

Instant alerts help you take action early



MANAGE

Personalized insights for better daily management



AI & Blood Sugar Monitoring

- **Content:**
 - Tracks glucose levels always
 - Predict Sudden Changes
 - Supports diabetic patients
 - Provides personalized insights



REAL-TIME
MONITORING

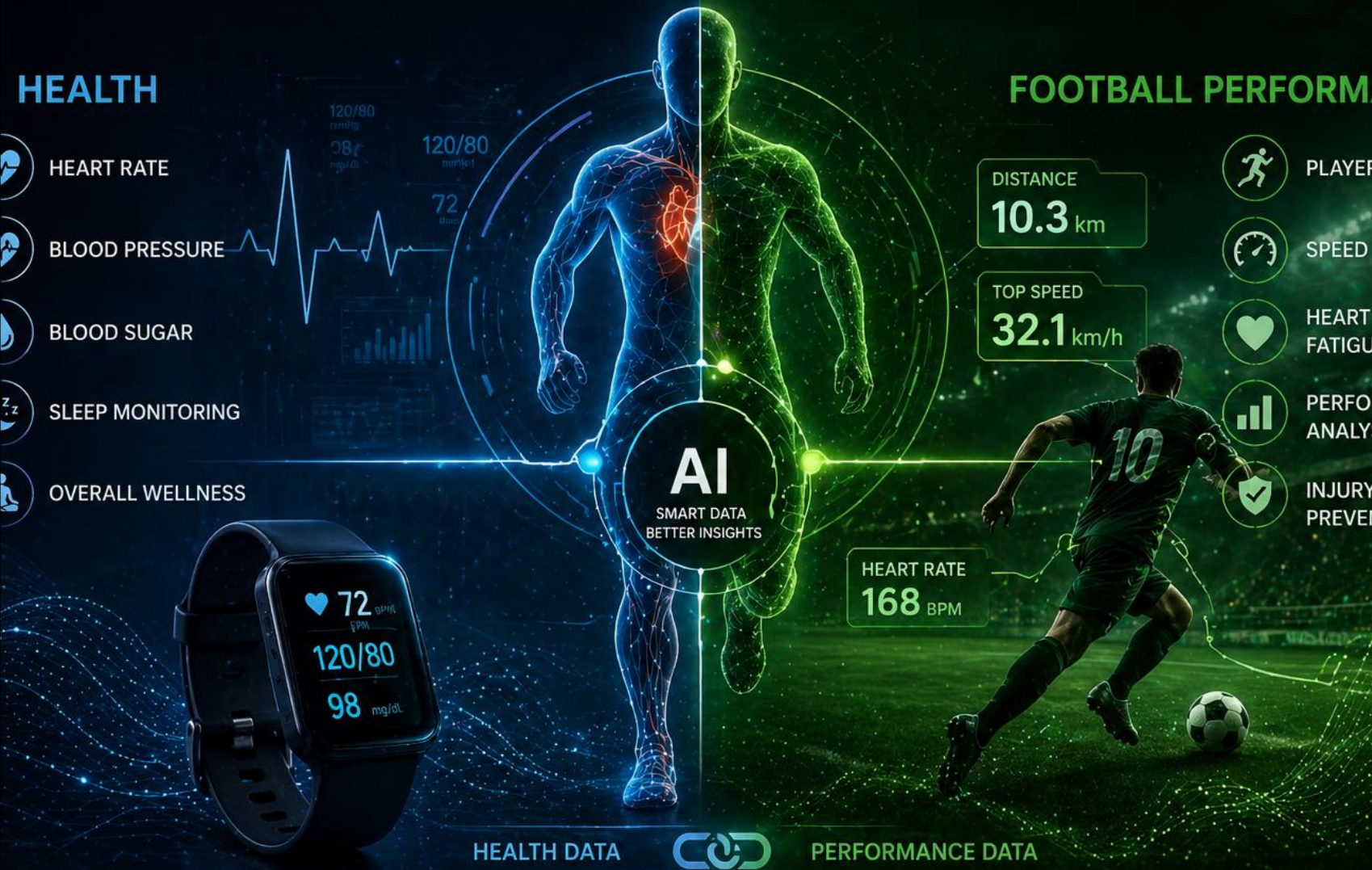


AI-POWERED
ANALYSIS



SMART
ALERTS

ONE TECHNOLOGY. TWO WORLDS.
BETTER HEALTH. BETTER PERFORMANCE.

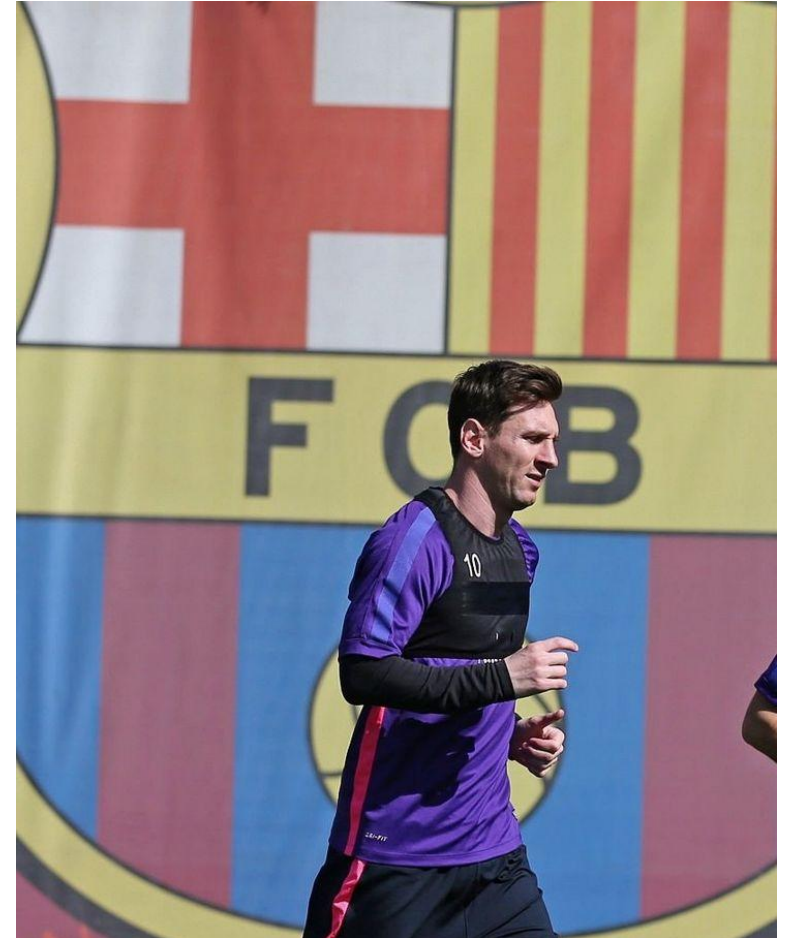


Connecting Health & Football

- Same wearable devices used in both fields
- Health data supports player performance
- Monitoring prevents injuries and fatigue
- AI transforms data into actionable insights

AI IN FOOTBALL PERFORMANCE

- Content :
- Tracks player movement and speed
- Monitors heart rate and fatigue
- Analyzes performance using data
- Helps improve training and prevent injuries

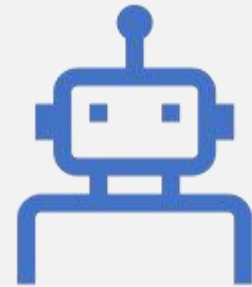


COM



AI will enhance prediction

Enhancing health and sports
performance



Real-time data is better

The future is intelligent and data-driven

END OF PRESENTATION

ANY QUESTIONS?...

THANK YOU

- **Presented by:**
- Mohamed Abdallah
- Marwan Zakaria
- Mahmoud Mohamed
- **Supervised by:**
- Dr. Samah El-Sakka

